

Principles of Microeconomics

Module 2: Gains from Trade

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What would happen if you had to make a sandwich completely on your own — grow the wheat, milk the cow, raise the chickens, and even harvest the salt? Watch this short video: *How to Make a \$1500 Sandwich in Only 6 Months*. It took one man 6 months and about \$1,500 to make just one sandwich. This story helps us think about a simple question: are we better off trying to do everything by ourselves, or working with others and trading

Video: “How to Make a \$1,500 Sandwich in Only 6 Months” (YouTube, *How To Make Everything*)

By the end of this module, you’ll be able to:

1. Explain and draw the Production Possibility Frontier (PPF).
2. Compare absolute and comparative advantage, and use comparative advantage to predict specialization.
3. Evaluate terms of trade and show how trade creates gains.

1 Understanding the Production Possibilities Frontier

Before we talk about trade, we need to understand what we can produce on our own. We’ll start by building a simple economic model that shows the limits of what’s possible when resources are scarce.

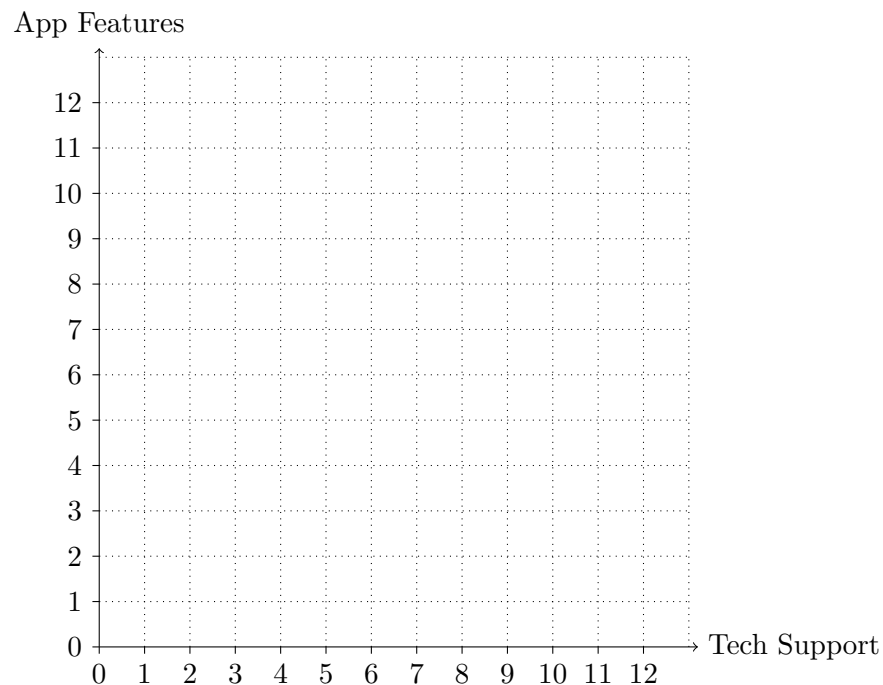
Key Definitions

- The **Production Possibilities Frontier (PPF)** is a model that shows all the possible combinations of maximum output a society can produce using all its available resources and technology.
- Because resources are limited, choosing one thing means giving up another. **Opportunity cost** is the value of the next best option you give up when making a choice. For example, imagine you have an economics problem set and a writing assignment. If you spend an hour solving the economics problem set, the opportunity cost is the writing you could have done in that time.

■ Let's say you manage a small tech company with a team of 6 workers. Each day, workers can be assigned to either: App Development, or Tech Support. Suppose all 6 workers are equally good at both tasks. Each worker can either:

- develop 2 app features per day, or
- resolve 1 tech support case per day.

- What is the production point when *all 6 workers* are assigned to **app development**?
- What is the production point when *all 6 workers* are assigned to **tech support**?
- Plot these two points and connect them to draw the PPF. (*Place Tech Support on the x-axis and App Features on the y-axis.*)
- What is the **opportunity cost** of resolving one tech support case?
- What is the slope of the PPF (*numerically*)?



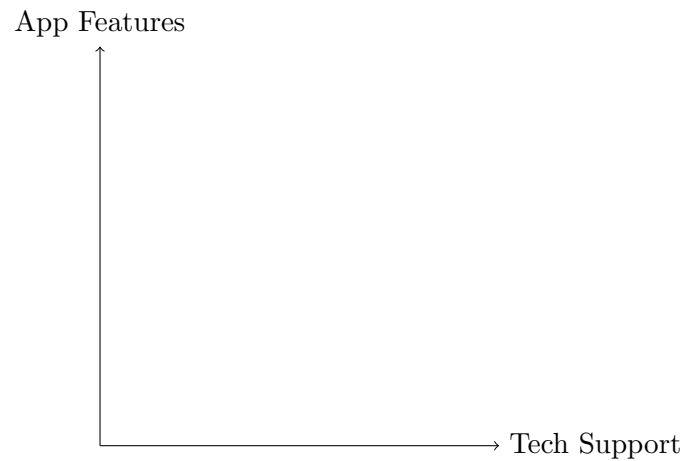
Let's talk about the slope:

- ► [Poll] What does the slope of the PPF represent (*economically*)?
- What does the **linear slope** of the PPF imply?

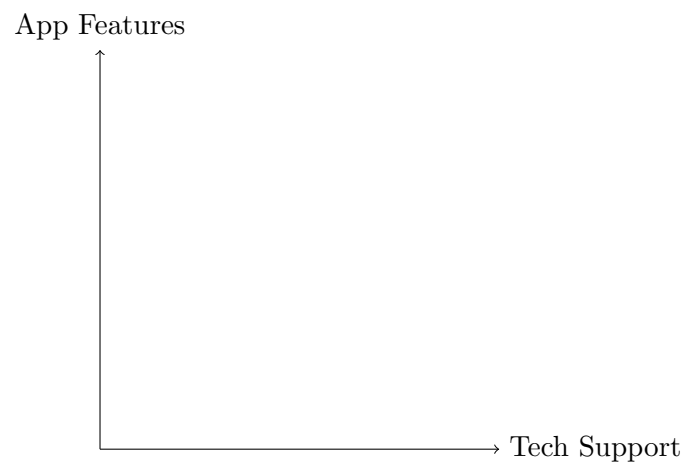
□ Let's discuss some points on the PPF graph in Activity 1.

So far, we have assumed that the company maximizes production given a fixed amount of resources and technology. But what if that changes?

- ► [Poll] The company introduces an AI chatbot that improves productivity in tech support. How would this affect the PPF?



- ► [Poll] Business is booming, so you hire 4 more workers. How would this affect the PPF?



□ In reality, a PPF is not always linear. You'll explore this further in Problem Set 1.

Key Concepts from the PPF

- **The Production Possibilities Frontier (PPF):** A model that shows all the possible combinations of two goods that can be produced efficiently using all available resources and current technology.
- **Efficiency:** A point is efficient if all available resources are fully used. All points **on** the PPF represent efficient combinations of the two goods.
- **Opportunity Cost and Slope:** Opportunity cost is what you give up when choosing one option instead of another. On a PPF, the **slope** represents the opportunity cost of the good on the horizontal (x) axis. In other words, the slope shows how much of the good on the vertical (y) axis must be given up to gain one more unit of the x -axis good.
- **Shifts of the PPF:** The entire PPF shifts outward when resources increase (e.g., more workers, more time) or when technology improves. It shifts inward when resources are reduced.
- **A Rotation of the PPF:** The PPF rotates when productivity or technology changes for only one good, while the other remains unchanged.
- **Increasing Opportunity Cost:** As you produce more of one good, you give up increasingly more of the other. This results in a PPF that is bowed out rather than a straight line.

2 Exploring the Benefits of Specialization and Trade

Once we know our limits, the next question is: will we be better off by specializing in what we are good at and trading? This section shows how society gains from specialization and trade. Let's go back to Andy, the YouTuber, who spent six months making a \$1,500 sandwich.

Key Definitions

- **Absolute Advantage:** A person or company has an *absolute advantage* in producing a good if they can produce more of it using the same amount of resources.
- **Comparative Advantage:** A person or company has a *comparative advantage* in producing a good if they can produce it at a lower opportunity cost than someone else.

2.1 Initial Productivity and Absolute Advantage

We will start with a very simple model using extreme assumptions. In this economy, there are two people, Andy and Bree, and two goods, Bread and Pickles. The table below shows how much each person can produce in an hour.

Table 1: Initial Productivity

Good	Andy	Bree
Bread (loaf)	1	3
Pickles (jar)	5	1

Use the table to answer the questions:

- (a) Who can produce more bread?
- (b) Who can produce more jars of pickles?

In economic terms, _____ has an **absolute advantage** in bread, while _____ has an **absolute advantage** in pickles.

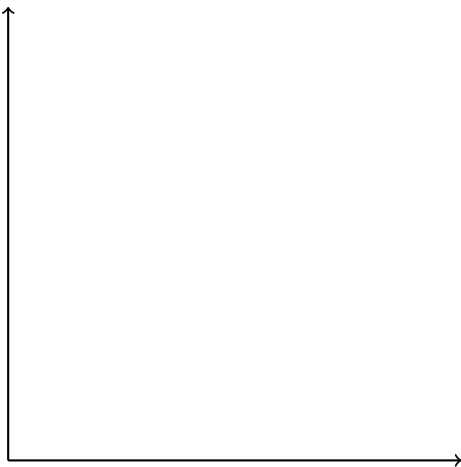
2.2 Maximum Production and the PPF

Suppose they each work 10 hours a day. Let's find their maximum possible output per day (if they specialize in only one good). This gives us each person's **Production Possibility Frontier (PPF)**.

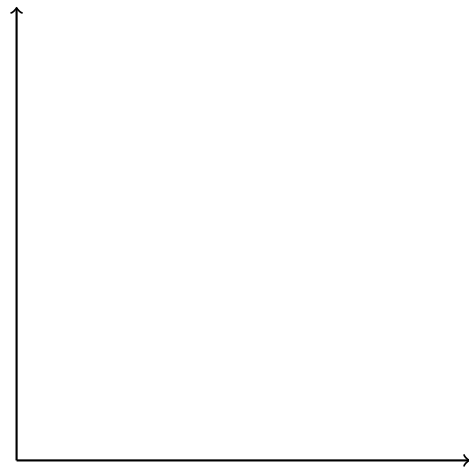
Table 2: Maximum Production

Good	Andy	Bree
Bread		
Pickles		

- (a) Calculate each person's maximum daily output if they spend all 10 hours producing only one good. Fill in the table above.
- (b) Draw each person's PPF. (Place Bread (B) on the x -axis and Pickles (P) on the y -axis.)
- (c) Write the equation of each PPF. (Express in terms of B and P .)



Andy's PPF



Bree's PPF

2.3 No Trade Scenario (Self-Sufficiency)

Table 3: Production/Consumption Without Trade

Good	Andy	Bree	Total (Economy)
Bread			
Pickles			

- (a) Suppose each person splits their time equally between bread and pickles. How much do they produce and consume? Fill in the table above.
- (b) Plot each production/consumption point on their PPF.

2.4 Comparative Advantage and Specialization

Rather than each trying to do a bit of everything, Andy and Bree can both gain by specializing. The question is: who should make what? Since we are not using money as a measure, the “cost” of producing something is what you have to give up, the **opportunity cost**.

Recall that if the production possibilities frontier shows two goods, say x and y , then

$$\text{Opportunity Cost of } x = \frac{\Delta y}{\Delta x}.$$

This tells us how many units of good y must be sacrificed to obtain one more unit of good x .

For example, if we let x = bread and y = pickles, then

$$\text{Opportunity Cost of 1 Bread} = \frac{\text{Pickles foregone}}{1 \text{ Bread}}.$$

Because we are almost always considering the cost of a single unit, the denominator “per bread” equals 1 and is usually left implicit. So we often just write

$$\text{Opp. cost of 1 Bread} = \text{some number (N) of pickles.}$$

with the “per bread” understood.

- (a) From Table 1, Andy can produce either 1 loaf of bread or 5 jars of pickles per hour. Bree can produce 3 loaves of bread or 1 jar of pickle per hour. Find the opportunity cost of making one loaf of bread for Andy and Bree.

Table 4: Opportunity Costs and Specialization

	Andy	Bree	Who Specializes?
Opp. Cost of 1 Loaf of Bread			
Opp. Cost of 1 Jar of Pickles			

- (b) Find the opportunity cost of making one jar of pickles for Andy and Bree.

⇒ In economic terms, _____ has a **comparative advantage** in bread, while _____ has a **comparative advantage** in pickles.

- (c) If they specialize, who will produce bread and who will produce pickles?
- (d) Calculate how much each produces under full specialization.

Table 5: Production With Trade

Good	Andy	Bree	Total (Economy)
Bread			
Pickles			

2.5 Terms of Trade and Gains from Trade

So far we've seen that specialization and trade can increase the economy's overall production level. But once two sides specialize, the next question is: at what rate should they trade with each other? This rate of exchange is called the *terms of trade*. With this in mind, we can examine when both sides will benefit.

Market for Bread:

- (a) Who is the seller of bread?
- (b) What is the minimum price, in terms of pickles per loaf of bread, that the seller would be willing to accept in trade?
- (c) Who is the buyer of bread?
- (d) What is the maximum price, in terms of pickles per loaf of bread, that the buyer would be willing to pay in trade?
- (e) What range of relative prices (in terms of pickles per loaf of bread) would both sides accept?

- (f) Suppose they agree on a trade price of 1 loaf of bread = 2 jars of pickles. If Andy buys 8 loaves of bread from Bree, fill in the table below:

Table 6: Consumption With Trade

	Andy	Bree
Bread		
Pickles		

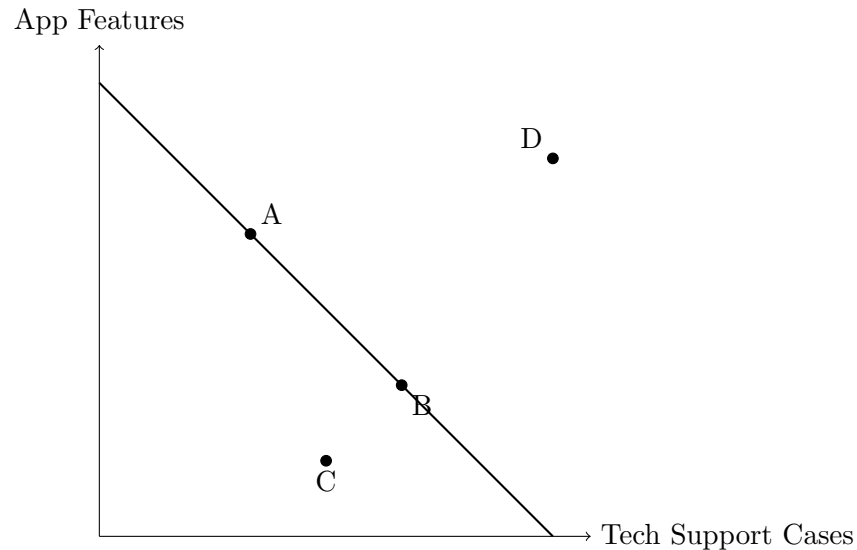
2.6 Analysis and Summary

- (a) Plot Andy and Bree's **consumption points after trade** on their PPFs.
- (b) Compare consumption before and after trade. Who is better off?
- (c) Summarize: How do specialization and voluntary trade benefit individuals and the economy?

- The economy _____
- Individuals _____

Activity 1: Understanding the Production Possibilities Frontier (Topic 1)

The graph below shows a production possibilities frontier for a small tech company that produces two services: **App Features** and **Tech Support Cases**.

**► [Poll]**

- (1) Which points are **efficient** combinations of app features and tech support?
- (2) Which point is **inefficient**?
- (3) Which point is **unattainable**? Why can't the company currently reach this point?
- (4) What would cause the PPF to shift outward? List one possible reason.
- (5) What would cause the PPF to pivot outward on just one axis (e.g., only for app development)?